

$$\begin{aligned}
 p(x) &= \int p(x, z) dz \\
 \mathcal{L}_p p(x) &= \mathcal{L}_p \int p(x, z) dz \\
 &= \mathcal{L}_p \int p(x, z) \cdot \frac{q_p(z|x)}{q_p(z|x)} dz \\
 &= \mathcal{L}_p \mathbb{E}_{z \sim q_p} \left[\frac{p(x, z)}{q_p(z|x)} \right] \\
 &\geq \mathbb{E}_{z \sim q_p} \left[\mathcal{L}_p \frac{p(x, z)}{q_p(z|x)} \right] \quad \leftarrow \text{Jensen's Inequality (Concave)} \\
 &= \mathbb{E}_{z \sim q_p} \left[\mathcal{L}_p \frac{p(x|z) p(z)}{q_p(z|x)} \right] \\
 &= \mathbb{E}_{z \sim q_p} [\mathcal{L}_p p(x|z)] - \mathbb{E}_{z \sim q_p} \left[\mathcal{L}_p \frac{q_p(z|x)}{p(z)} \right] \\
 &= \underbrace{\mathbb{E}_{z \sim q_p} [\mathcal{L}_p p(x|z)]}_{\text{Reconstruction}} - \underbrace{D_{KL}(q_p(z|x) \| p(z))}_{\text{Matching Term}}
 \end{aligned}$$

$$\mathcal{L} = \underbrace{-\mathbb{E}_{z \sim q_p} [\mathcal{L}_p p(x|z)]}_{\text{Reconstruction}} + \underbrace{D_{KL}(q_p(z|x) \| p(z))}_{\text{Matching Term}}$$

$$p_\theta(x|z) = \mathcal{N}(x; \mu_\theta, \Sigma)$$

$$q_\theta(z|x) = \mathcal{N}(z; \mu_\theta, \sigma^2 I) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(z-\mu)^2}{2\sigma^2}} \quad p(z) = \mathcal{N}(0, I)$$

① Matching Term

$$\begin{aligned}
 D_{KL}(q_\theta(z|x) \| p(z)) &= \mathbb{E}_{z \sim q_\theta} [\mathcal{L}_p q_\theta(z|x) - \mathcal{L}_p p(z)] \\
 &= \mathbb{E}_{z \sim q_\theta} [\mathcal{L}_p q_\theta(z|x)] - \mathbb{E}_{z \sim q_\theta} [\mathcal{L}_p p(z)] \\
 \text{Note that } \mathcal{L}_p \mathcal{N}(\mu, \sigma^2) &= \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(z-\mu)^2}{2\sigma^2}} \\
 &\rightarrow \left(\mathcal{L}_p \frac{1}{\sqrt{2\pi}\sigma} + \mathbb{E} \left[-\frac{(z-\mu)^2}{2\sigma^2} \right] \right) - \left(\mathcal{L}_p \frac{1}{\sqrt{2\pi}} + \mathbb{E} \left[-\frac{z^2}{2} \right] \right) \\
 \text{Note } \mathbb{E}[(z-\mu)^2] &= \sigma^2, \quad \mathbb{E}[z^2] = \text{Var}(z) + \mathbb{E}[z]^2 = \sigma^2 + \mu^2 \\
 &\rightarrow \mathcal{L}_p \frac{1}{\sqrt{2\pi}\sigma} - \frac{1}{2} - \mathcal{L}_p \frac{1}{\sqrt{2\pi}} + \frac{(\sigma^2 + \mu^2)}{2} \\
 &= -\mathcal{L}_p \sigma - \frac{1}{2} + \frac{1}{2}(\sigma^2 + \mu^2) \\
 &= -\frac{1}{2} \left(1 + \log \sigma^2 - \sigma^2 - \mu^2 \right)
 \end{aligned}$$

$$\mathcal{J} = -\mathbb{E}_{z \sim q_\theta} [\log p(z|x)] + D_{KL}(q_\theta(z|x) \| p(z))$$

Reconstruction Matching Term

$$p_\theta(z|x) = \mathcal{N}(z; \mu_\theta, \sigma^2 \mathbf{I})$$

$$q_\theta(z|x) = \mathcal{N}(z; \mu_\theta, \sigma^2 \mathbf{I}) \quad p(z) = \mathcal{N}(0, \mathbf{I})$$

① Matching Term

$$D_{KL}(q_\theta(z|x) \| p(z))$$

$$= \mathbb{E}_{z \sim q_\theta} [\log q_\theta(z|x) - \log p(z)]$$

$$= \mathbb{E}_{z \sim q_\theta} [\log q_\theta(z|x)] - \mathbb{E}_{z \sim q_\theta} [\log p(z)]$$

$$\text{Use the } \mathcal{N}(\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(z-\mu)^2}{2\sigma^2}}$$

$$\rightarrow \left(\log \frac{1}{\sqrt{2\pi}\sigma} + \mathbb{E} \left[-\frac{(z-\mu)^2}{2\sigma^2} \right] \right) - \left(\log \frac{1}{\sqrt{2\pi}} + \mathbb{E} \left[-\frac{z^2}{2} \right] \right)$$

$$\text{Use } \mathbb{E}[(z-\mu)^2] = \sigma^2, \quad \mathbb{E}[z^2] = \text{Var}(z) + \mathbb{E}[z]^2 = \sigma^2 + \mu^2$$

$$\rightarrow \log \frac{1}{\sqrt{2\pi}\sigma} - \frac{1}{2} - \log \frac{1}{\sqrt{2\pi}} + \frac{(\sigma^2 + \mu^2)}{2}$$

$$= -\log \sigma - \frac{1}{2} + \frac{1}{2}(\sigma^2 + \mu^2)$$

$$= -\frac{1}{2} \left(1 + \log \sigma^2 - \sigma^2 - \mu^2 \right)$$

② Reconstruction Term

$$\mathbb{E}_{z \sim q_\theta} [\log p_\theta(z|x)]$$

$$\simeq \frac{1}{N} \sum \log p_\theta(z|x)$$

$$= \text{MSE}$$

→ name code

1. $q_\theta(z|x) = \mathcal{N}(\mu_\theta(x), \sigma_\theta^2(x) \mathbf{I}) \rightarrow$ does not depend on integral over z !

Thus, $D_{KL}(q_\theta(z|x) \| p(z)) = \int q_\theta(z|x) \log \frac{q_\theta(z|x)}{p(z)} dz$ is tractable

2. $p_\theta(z|x) = \mathcal{N}(\mu_\theta(z), \sigma^2 \mathbf{I}) \rightarrow$ does depend on z !

Thus, $\mathbb{E}_{z \sim q_\theta} [\log p_\theta(z|x)] = \int q_\theta(z|x) \cdot \log p_\theta(z|x) dz$ is intractable